VAISHNAV K

SENIOR DATA ENGINEER

Apache Spark – Big Data – SQL - AWS – Databricks – ETL/ELT

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PROFESSIONAL SUMMARY

- Senior Data Engineer with **5 years** of experience designing, developing, and supporting **high-performance data engineering solutions**.
- **Led** Spark and Databricks delivery for 2 major banks, **Citi Bank** and **Bank of America**, owning design, release, support, and mentoring.
- Strong experience with 8+ core technologies including **Apache Spark, PySpark, Spark SQL, Databricks, AWS, SQL, Hadoop, and Hive.**
- Delivered 30+ fault-tolerant **batch and streaming data pipelines** supporting terabyte-scale workloads and distributed processing.
- Skilled in Delta Lake, **Unity Catalog, Kafka, Structured Streaming, Glue, Lambda,** and **Athena** across 2 **cloud environments**.
- Hands-on experience with 5+ practices like **data integration, workflow orchestration, monitoring, AI engineering,** and **RBAC security**.

KEY SKILLS

- **4+ years Banking Data Engineering** across financial services, scalable data platforms, and production workloads.
- **30+ Spark Performance Optimizations** involving query tuning, AQE, data skew, schema evolution, and AI-leveraged development.
- **10+ Legacy System Modernization & Cloud Migration** initiatives facilitating extensible platforms, Terraform, IaC, and Lakehouse.
- **3 Major Banking Domain Data Projects** spanning trade execution, fraud transaction, and customer attribute pipelines at Citi and BofA.
- **Terabyte-scale Distributed Data Processing** across 30+ production-ready live pipelines, supporting high-throughput batch workloads.
- **Stakeholder Communication & Cross-Functional Delivery** across strategic efforts with architects, analysts, QA, and 5+ business teams.
- **15+ Data Pipeline Monitoring & Incident Management activities** maintaining data ingestion, reliability, and SLA management.
- **Data Security & Access Control activities across banking platforms** supporting RBAC governance, PII masking, and GDPR compliance.
- **10+ Real-Time Streaming Data Pipelines** using Kafka, Kinesis, and Structured Streaming for 24/7 banking analytics.
- **Data Governance & Quality** practices on 10+ enterprise assets using Unity Catalog, RBAC, MDM, data lineage, and Great Expectations.
- **10+ Major Production Releases** & Release Management activities following CI/CD, testing, deployment, and environment validation.
- **15+ Apache Spark & Databricks Development projects** with flexible workflows, distributed processing, and cloud-native ecosystem.

TECHNICAL SKILLS

- **Big Data:** 9+ competencies | Apache Spark | PySpark | Spark SQL | Hadoop | HDFS | Hive | Kafka | Structured Streaming | Sqoop.
- **Cloud:** 10+ technologies | AWS | S3 | EMR | EC2 | IAM | Snowflake | Redshift | GCP | Glue | Lambda | Athena | Kinesis | BigQuery.
- **Lakehouse**: 6+ competencies | Databricks | Delta Lake | Unity Catalog | Medallion Architecture | Delta Live Tables (DLT) | Workflows.
- **Data Eng.:** ETL/ELT | 30+ Batch & Stream Pipelines | Data Warehousing | CDC | SCD Type 2 | dbt | Data Partitioning & Masking.
- **Programming & AI:** 9+ Skills | SQL | Python | Scala | Java | Shell Scripting | Generative AI | LLMs | Prompt Engineering | GitHub Copilot.
- **Databases & Data Formats:** 8+ DBs & formats | PostgreSQL | SQL Server | Parquet | Avro | ORC | Iceberg | DynamoDB | Cassandra.
- **Orchestration, DevOps & Tools**: 9+ tools | Autosys | Airflow | Git | Jenkins | Ansible | uDeploy | Terraform | Kubernetes | Docker.

KEY PROJECTS

- **Order Execution Trade Flow – Databricks Integration (Citi Bank):** Migrated 15+ **legacy Apache Spark pipelines** to Databricks, utilizing AWS S3, GitHub Copilot & Agents to modernize banking data workloads across 2 cloud environments. Applied Spark performance and cluster optimization, scalable ELT and data architecture practices, improving efficiency and reducing aggregation time by ~85%.
- **Behavioral Biometrics – Data Pipeline Migration (Bank of America):** Developed 10+ **data streaming ingestion** pipelines using Spark Structured Streaming for near-real-time data handling and analytics across multiple data sources. Established high-throughput data integration, stream processing, data transformation, framework and monitoring workflows, supporting 24/7 downstream analytics.
- **Fraud Transactional and Customer Attributes – Hybrid Flow (Bank of America):** Designed a **hybrid Hadoop and AWS S3** data architecture for storage, Delta Lake, and analytics of ~12 TB of attribute data across 3 environments. Engineered data integration, distributed data processing, and transformation across on-premises and cloud environments, supporting 10+ downstream workflows.

WORK EXPERIENCE

Senior Data Engineer – LTIMindtree/LTM (Client: Citi Bank) November 2024 – Present

- Led the design, development, and deployment of **15+ production-grade Spark and Databricks clusters** across cloud environments using Terraform and AWS CloudFormation to support enterprise banking applications **processing 10+ TB of data.**
- **Owned end-to-end delivery across 6+ stages**, incorporating **Data Modeling (Dimensional Modeling, Star Schema, SCD Type 2)** designs for requirements gathering, design, development, data warehousing, testing, deployment, and production support.
- Designed **AWS S3 data lake architecture** and integrated **3+ downstream analytics platforms** including **Snowflake**, **Redshift,** and **Amazon Athena** using **AI-Assisted Development**. Tuned Scala/PySpark jobs using partitioning, caching, broadcast joins, AQE, and predicate pushdown, **cutting runtime by 45%** across large-scale data processing workloads and production data pipelines.
- **Automated** scheduling and dependencies with **Airflow DAGs**, orchestrated ELT transformations with **dbt**, and implemented data quality frameworks like **Great Expectations** across 15+ operationalized pipelines to improve stability and observability.
- Leveraged **GitHub Copilot, AI agents, and AI data engineering** across 10+ projects, supporting code development, Spark troubleshooting, SQL optimization, documentation, code review, test case generation, and productivity.
- Integrated **Delta Lake**, **Unity Catalog,** and **Medallion Architecture data management**, establishing **RBAC (Role-Based Access Control)** and **Master Data Management (MDM)** protocols across 10+ enterprise data assets to support secure governance.
- Managed 10+ event-driven data workflows using **AWS Glue** and **AWS Lambda**, incorporating CDC, metadata, and data lineage.
- Collaborated with **5+ cross-functional teams**, including architects, developers, analysts, QA, post-deployment teams, and business stakeholders, to deliver **enterprise-grade data solutions** across complex banking environments.
- **Mentored** 3+ junior engineers, performed code reviews across 15+ pipelines, and drove architecture and technical design.

Big Data Developer – BA Continuum Pvt Ltd (Client: Bank of America) April 2022 – October 2024

- Developed and maintained **30+ Spark, Hive, and Hadoop workflows,** building daily batch and near-real-time Structured Streaming CDC pipelines using Kafka, **Amazon Kinesis**, and Spark to process terabyte-scale data.
- Migrated **10+ legacy Hadoop-based workloads** to **cloud-based architectures** on **AWS and GCP (**leveraging **Google BigQuery),** containerizing applications with **Docker** and orchestrating with **Kubernetes (K8s)** to improve scalability.
- Configured Spark transformations, revamped Hive queries across **20+ datasets**, and automated 8+ releases via **GitHub Actions**.
- Coordinated with **5+ business stakeholders, analysts, and engineering teams** in an Agile environment to deliver scalable data solutions **across enterprise banking projects** and **production environments**.
- Supported 7+ high-availability data pipelines integrating structured and NoSQL databases like **PostgreSQL** and **Amazon DynamoDB**, ensuring reliable enterprise data delivery.

ETL Developer Intern – Seyasoft Technologies June 2021 – April 2022

- Integrated data from 10+ relational, file-based, and streaming sources using **Data Vault** methodologies, schema evolution, and ETL workflows for enterprise customer applications across multiple downstream systems.
- Optimized 15+ Spark jobs and Hive queries using partitioning, caching, and query tuning techniques, **reducing processing time by 65%** while improving business-critical performance across terabyte-scale workloads.

EDUCATION

- MASTER'S DEGREE | **Business Administration (Business Analytics)** | Amity University (Pursuing) **2026 – 2028**
- BACHELOR'S DEGREE | **Computer Science Engineering (AI&ML)** | SRM University – **CGPA: 9.9** **2018 – 2022**
- HIGH SCHOOL DIPLOMA | **Computer Science (CBSE)** | Amrita Vidyalayam **2016 – 2018**

CERTIFICATIONS

- Databricks Certified **Data Engineer Professional** **2025**
- Databricks Certified **Data Engineer Associate** **2025**
- Databricks Certified **Associate Developer for Apache Spark 3.0** **2025**